

Clinical Outcomes of Rotator Cuff Tear Repair in Patients ≤ 45 Years of Age

A Retrospective Analysis of 108 Patients

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Background: In the general population, the prevalence of rotator cuff tears (RCTs) is approximately 20.7%, with good clinical outcomes typically observed after rotator cuff repair (RCR). This prevalence is lower among patients ≤ 45 years of age. However, younger patients often have higher expectations after RCR. Despite this, few studies have specifically focused on this population.

Purpose/Hypothesis: The purpose of this study was to evaluate the clinical outcomes of patients ≤ 45 years of age after full-thickness RCR. It was hypothesized that RCR improves pain score, strength, and clinical score significantly in this age group.

Study Design: Case series; Level of evidence, 4.

Methods: The authors retrospectively analyzed the records of patients aged ≤ 45 years at the time of surgery who underwent arthroscopic repair of a full-thickness RCT from 2011 to 2022 in 10 institutions by senior shoulder surgeons. Exclusion criteria were partial tears, revision surgery, and follow-up < 1 year. Clinical follow-ups were conducted at 12, 24, and 120 months postoperatively, including evaluations of pain using the visual analog scale (VAS) score, range of motion, strength in abduction, and Constant-Murley score assessments. Additionally, the complication rate and the duration of preoperative symptoms were recorded.

Results: The authors retrospectively included 108 shoulders in 108 patients ≤ 45 years of age at the time of surgery who underwent full-thickness arthroscopic repair by senior fellow-trained shoulder surgeons. A double-row repair was performed in 40% of cases; otherwise, a single-row repair was performed for posterosuperior RCTs. The mean age was 40.5 years, and 66% were men, with a mean follow-up of 2.4 years. Tears were traumatic in 41% of cases. Patients were employed in 97% of cases: 19% were office workers, 46% were light manual workers, and 32% were heavy workers. The VAS score, forward elevation, external and internal rotation, Constant score, and strength had all significantly improved at 1 year postoperatively. The mean Constant score gain was 30.0 in patients with a traumatic tear and 21.0 in patients with a degenerative tear ($P = .001$). The increase between preoperative and postoperative Constant scores was higher in patients with symptoms for < 6 months at the time of surgery than in patients with symptoms for ≥ 6 months at the time of surgery ($P = .023$). Few (8.3%) complications were observed; most of them (8 patients) were postoperative stiffness that needed a specific treatment, and 1 patient had a symptomatic retear. In multivariate analysis, shoulder trauma was associated with higher postoperative Constant scores ($P = .0042$). Work-related tear was associated with lower Constant scores at 1 year ($P = .0301$).

Conclusion: Patients ≤ 45 years of age with RCT generally have a good clinical outcome with improvement of pain and Constant score after arthroscopic repair. Degenerative tears and manual work are associated with poorer but acceptable outcomes.

Keywords: rotator cuff repair; patients ≤ 45 years of age; traumatic versus degenerative tear

better healing potential with better tendon quality and vascular supply than older patients.¹ In the general population, good clinical outcomes have been shown after rotator cuff repair (RCR).^{2,8,15,23,25} However, few studies have evaluated the results of younger patients. Indeed, young patients have a higher level of activity and higher physical demand and expect to return to sport activities and work occupation at the same level as before the RCT. For this reason, satisfaction results imply a higher clinical outcome than those for older patients. Moreover, the time lost from work is a major public health issue and has a significant impact on society. Therefore, it is important to understand the results and evolution of RCR in this age group.

The purpose of this study was to evaluate the characteristics and outcome of younger patients after full-thickness RCR. We hypothesized that RCR would significantly improve pain score, strength, and clinical score in this age group.

METHODS

Study Design and Patient Characteristics

Institutional review board approval (COS-RGDS-2021-06-002-HARDY-A) was granted for this retrospective cohort study. A retrospective analysis of prospectively collected data was performed. Patients aged ≤ 45 years at the time of surgery who underwent arthroscopic repair of a full-thickness RCT from 2011 to 2022 in 10 institutions by senior shoulder surgeons were considered for inclusion. Exclusion criteria were partial tears, revision surgery, and follow-up < 1 year. All patients provided informed consent. This study was based on a multicenter database compiled for a symposium of the French Society of Arthroscopy (Société Francophone d'Arthroscopie), in which participating surgeons retrospectively contributed data from patients who met the predefined inclusion criteria. Only eligible cases were submitted.

Preoperative Assessment

The preoperative evaluation involved both clinical and imaging assessments. Clinically, range of motion (ROM), abduction strength, and the Constant-Murley score were recorded. Tendon tear size based on the Patte classification was assessed using magnetic resonance imaging (MRI) scans, and fatty infiltration of the rotator cuff muscles based on the Goutallier system was assessed using MRI scans⁹ or computed tomography arthrograms.¹¹ The level of work was assessed preoperatively (light manual worker, heavy manual worker, and office worker).

Although this was a retrospective multicenter study without a unified treatment protocol, surgical indications followed general clinical practice standards in France. In patients ≤ 45 years of age with traumatic, full-thickness (transfixing) RCTs, surgery was typically indicated systematically. A traumatic tear was defined clinically by the occurrence of a distinct traumatic event followed by the immediate onset of shoulder pain and/or functional limitation. The Collin classification was used to describe the topographic pattern of multiple-tendon tears.⁶ For non-traumatic tears, surgery was considered after the failure of nonoperative management, which generally included several months of physiotherapy and/or corticosteroid injections. The decision for surgery was ultimately left to the treating surgeon's judgment based on symptom persistence and functional impairment.

Surgical Technique and Postoperative Care

All surgical interventions were performed by one of the senior shoulder surgeons (M.C., P.A., M.A., L.B., A.H., Y.B., X.O., C.E.T., A.G.) with the patient in the beach-chair position under general anesthesia after administration of a preoperative interscalene nerve block. All patients were operated on with an arthroscopic procedure. Either a double-row repair or a single-row repair was performed for posterosuperior RCTs, depending on the surgeon's preference.

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Final revision submitted May 20, 2026; accepted May 24, 2026.

One or more of the authors has declared the following potential conflict of interest or source of funding: A.H. has received consulting fees from Arthrex. M.A. has received consulting fees, royalties, and speaker fees from FX Shoulder Solutions and CONMED; and consulting fees from Pixee. M.C. is a co-founder of and shareholder in EasyMedStat. P.A. has received consulting fees from J&J, Exactech, and Medacta. L.B. has received consulting fees from Stryker and Enovis. Y.B. has received consulting fees and royalties from Medacta. X.O. has received consulting fees from Stryker Arthroscopy. A.G. reports consulting or advisory relationships with Arthrex, CONMED Corporation, FH, Move-UP, Vims, and Stryker. GCS RAMSAY SANTE funds the scientific activity at the Santy center.

Ethical approval for this study was obtained from Conseil d'Orientation Scientifique Ramsay Santé (COS-RGDS-2021-06-002-HARDY-A).

Additional arthroscopic procedures included acromioplasty, biceps tenodesis or tenotomy, and distal clavicle resection as needed. When the long head of the biceps brachii tendon was found to be pathological, tenodesis or tenotomy was performed, regardless of its position within the bicipital groove (whether in place, subluxated, or dislocated).

Postoperatively, the shoulder was immobilized for 6 weeks with a pillow or sling (depending on the center) in 20° of abduction. Rehabilitation began within the first week after surgery with passive stretching exercises to restore ROM. Hydrotherapy was introduced when skin healing allowed. Daily activities involving the operated shoulder were permitted after 3 months, with sports and strenuous tasks gradually resumed after 6 months.

Postoperative Clinical Evaluation

Clinical follow-ups were conducted at 12, 24, and 120 months postoperatively, including evaluations of pain using the visual analog scale (VAS) score, ROM (external rotation was measured with the arm at the side, and internal rotation was assessed based on the highest vertebral level reached by the thumb behind the back), strength in abduction, and Constant score assessments. This reflects our standard practice for patient follow-up. However, some patients did not reach the 120-month visit, as the last surgeries were performed in 2022. Additionally, the complication rate and the duration of preoperative symptoms were recorded. For patients unavailable to visit an outpatient clinic, telephone evaluation was conducted and a questionnaire completed.

Statistical Analysis

Numeric data are expressed as mean with standard deviation and discrete data as frequency with percentage. Normality was assessed using the Shapiro-Wilk test and homoscedasticity using the Levene test. Differences between preoperative and postoperative outcomes were assessed with the Wilcoxon signed-rank test. Groups of patients were compared using the Mann-Whitney test or Kruskal-Wallis test. If the null hypothesis of the Kruskal-Wallis test was rejected, post hoc pairwise analyses were performed with the Dunn-Bonferroni test. The alpha risk was set to 5% ($\alpha = .05$). A multivariate linear regression was performed to assess the relation between the 1-year Constant score and the explanatory variables: shoulder trauma, age at inclusion (years), work-related tear, occupation, and multiple-tendon tear. Data were checked for multicollinearity using the Belsley-Kuh-Welsch technique. Heteroscedasticity and normality of residuals were assessed using the Breusch-Pagan test and the Shapiro-Wilk test, respectively. Shoulders with missing data were excluded from the analysis. The Newey-West correction for heteroscedasticity was applied. Statistical analysis was performed using EasyMedStat (Version 3.24; www.easymedstat.com). Light manual workers and heavy workers were categorized based on

TABLE 1
Baseline Characteristics

	Value
Age at surgery, y	
Mean $\pm$ SD	40.5 $\pm$ 6.9
Minimum, maximum	12, 45
Male sex, %	66
Dominant side, %	73.5
Traumatic tear, %	41
Duration of symptoms before surgery, mo	
Mean $\pm$ SD	23.9 $\pm$ 30.1
Minimum, maximum	0, 190
Follow-up, y	
Mean $\pm$ SD	2.4 $\pm$ 1.8
Minimum, maximum	1, 10.6
Type of occupation, %	
Office worker	19
Light manual worker	46
Heavy manual worker	32
Tear pattern	
Isolated tendon tear, %	52
Isolated supraspinatus	86
Isolated subscapularis	14
Multiple-tendon tear, % ^a	48
Collin type A	15
Collin type B	0
Collin type C	23
Collin type D	62
Collin type E	0
Tendon retraction, %	
Patte stage 1	85
Patte stage 2	13
Patte stage 3	2
Fatty degeneration, %	
Goutallier grade 0	55
Goutallier grade 1	42
Goutallier grade 2	3

^aCollin classification for massive chronic rotator cuff tears.⁶

previous publications. Light manual workers included caregivers, electricians, and gardeners. Heavy manual workers included construction workers and truck drivers. The minimal clinically important difference (MCID) was analyzed based on the publication of Kukkonen et al.¹⁵

RESULTS

We retrospectively included 108 patients. Descriptive data and tear characteristics are presented in Table 1. Patients were employed in 97% of cases: 19% were office workers, 46% were light manual workers, and 32% were heavy workers.

Tear Patterns and Surgical Data

Tear patterns and tendon findings are presented in Table 1. The tear was traumatic in 41% of cases and associated with a dislocation in 7%.

TABLE 2
Preoperative and Postoperative Clinical Scores^a

	Preoperative Score	Postoperative Score	P Value
Pain (VAS score)	6.6 ± 1.1	2.1 ± 2.3	<.001
Forward elevation, deg	153.5 ± 34.9	168.6 ± 20.9	<.001
External rotation, deg	46.9 ± 21.3	58.5 ± 19.0	<.001
Internal rotation, points ^b	6.8 ± 2.2	8.7 ± 1.8	<.001
Constant score	56.2 ± 13.6	80.8 ± 16.1	<.001
Strength, points	9.1 ± 5.6	16.1 ± 7.2	<.001

^aData are presented as mean ± SD. VAS, visual analog scale.

^bValues for internal rotation correspond to the points attributed for internal rotation in the Constant score.

Tears involved an isolated tendon in 56 (52%) patients: 48 (86%) had a supraspinatus tear and 8 (14%) had a subscapularis tear. Multiple tendons were involved in 52 (48%) patients; subtypes are listed in Table 1. Overall, 98% of the shoulders had no or slight muscle atrophy on MRI.

A double-row repair was performed in 40% of cases; otherwise, a single-row repair was performed for posterosuperior RCTs. Additional surgical procedures included acromioplasty (n = 74, 69%), biceps tenodesis (n = 47; 44%), biceps tenotomy (n = 25; 23%), and distal clavicle resection (n = 5; 4.95%). In 1 case, the long head of the biceps was ruptured at the time of surgery and was not repaired.

Postoperative rehabilitation involved either a sling (49%) or an abduction pillow (51%). Early rehabilitation, within the first week, was started in 64% of patients.

Clinical Results

Clinical score outcomes are presented in Table 2. The VAS score, forward elevation, external rotation, internal rotation, the Constant score, and strength all showed significant improvement postoperatively. The mean Constant score gain was 24.4 ± 18.1 .

The median preoperative and postoperative Constant scores were 57.0 (IQR, 14.0) and 83.5 (IQR, 22.25), respectively. The median difference was -23.0 (IQR, 19.0; 95% CI, -27.0 to -21.0; $P < .001$) (Figure 1). Overall, 82% of patients had an increase in Constant score of ≥ 10.4 points, which exceeded the MCID¹⁰ (Figure 2).

Work-Related Tear

The median postoperative Constant scores were 80.0 (IQR, 22.0), 87.5 (IQR, 23.25), and 89.0 (IQR 9.0) for light manual workers, heavy manual workers, and office workers, respectively ($P = .015$) (Figure 3, Table 3). Pairwise analyses revealed differences for light manual workers versus heavy manual workers ($P = .015$) and office workers versus light manual workers ($P = .006$). Overall, 96% of office workers had an increase in Constant score that was greater than or equal to the MCID ($P = .039$). The median

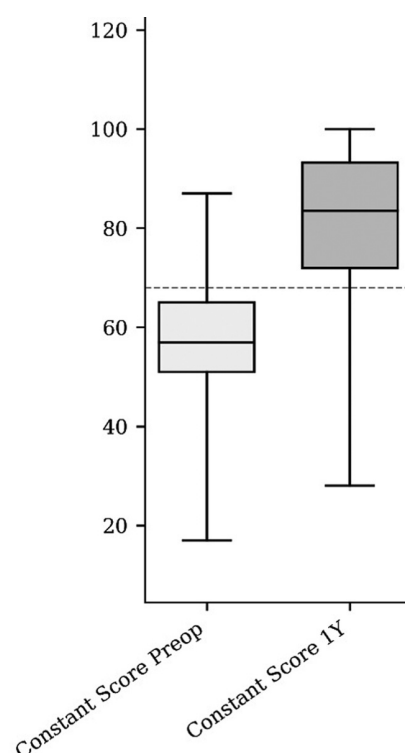


Figure 1. Preoperative and 1-year follow-up Constant score box plots.

postoperative Constant score gains were 20.0 (IQR, 20.25), 27.0 (IQR, 27.0), and 23.0 (IQR, 10.0) for light manual workers, heavy manual workers, and office workers, respectively ($P = .203$).

Traumatic Versus Degenerative Tears

The median postoperative Constant score was 90.5 (IQR, 14.25) in patients with a traumatic tear and 80.0 (IQR, 22.0) in patients with a degenerative tear ($P < .001$) (Figure 4, Table 3). The median postoperative Constant score gain was 30.0 (IQR, 17.0) in patients with a traumatic

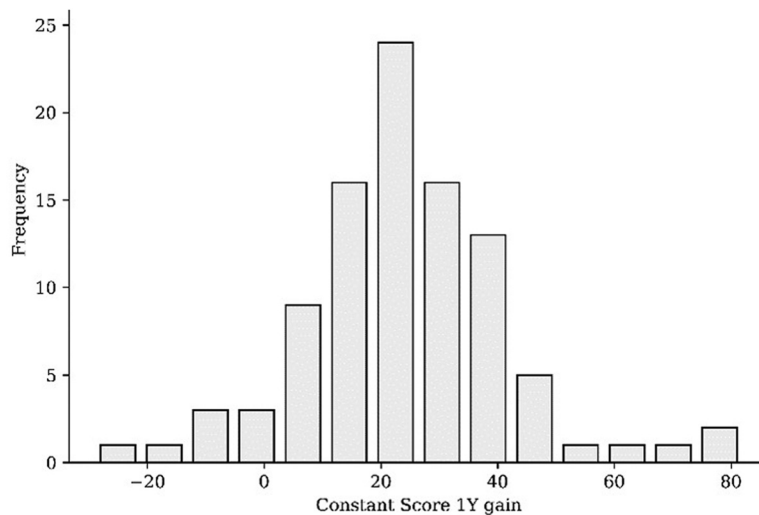


Figure 2. Constant score gain at 1-year follow-up.

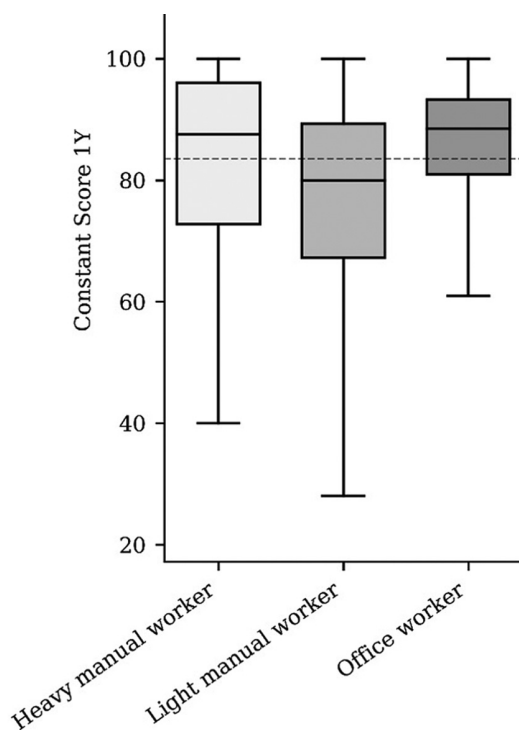


Figure 3. One-year follow-up Constant score box plot by occupation.

tear and 21.0 (IQR 14.5) in patients with a degenerative tear ($P = .001$) (Figure 5). Overall, 95% of patients with a traumatic tear had an increase in Constant score that was higher than the MCID, compared to 76% of patients with a degenerative tear ($P = .02$).

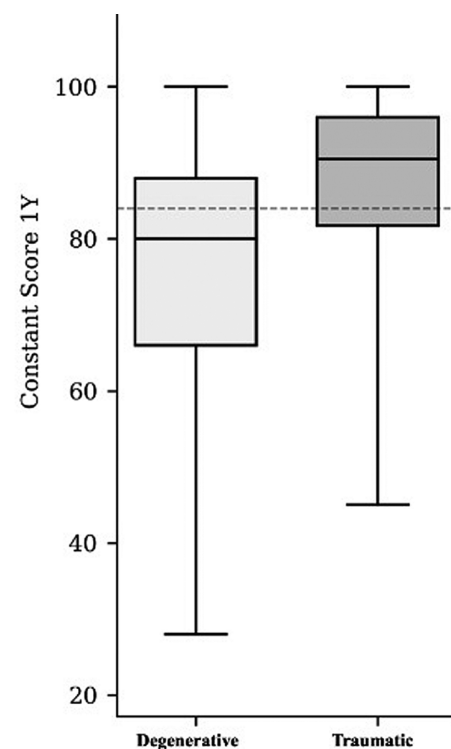


Figure 4. One-year follow-up Constant score box plot by tear type.

In our series, patients with a traumatic tear were slightly younger (38 ± 9 years vs 42 ± 5 years old; $P = .03$), were operated on sooner (17 vs 30 months after beginning of symptoms; $P < .001$), had a better postoperative Constant score (88 ± 12 points vs 76 ± 17 points;

TABLE 3
One-Year Follow-up Constant Score by Subgroup^a

	Score	P Value
Tear type		
Traumatic tear	88 ± 12	.001
Degenerative tear	76 ± 17	
Symptom duration		
Symptoms for ≥6 mo at time of surgery	80 ± 15	.024
Symptoms for <6 mo at time of surgery	87 ± 17	
Occupation type		
Office worker	87 ± 9	.016
Light manual worker	76 ± 17	
Heavy manual worker	83 ± 17	

^aData are presented as mean ± SD.

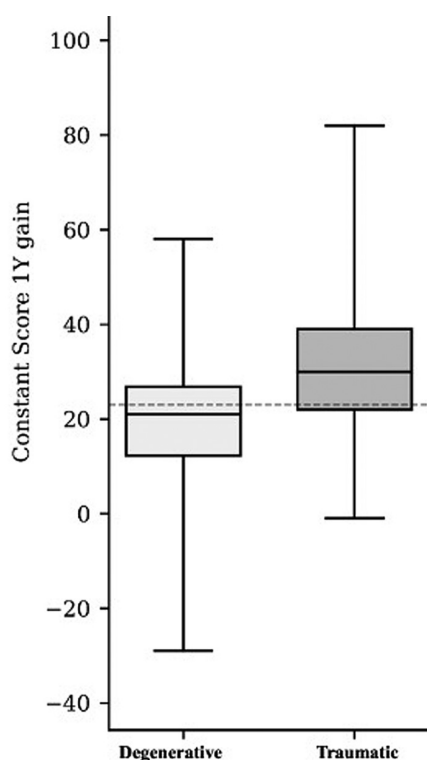


Figure 5. One-year follow-up Constant score gain box plot by tear type.

$P < .001$), and had a higher increase between preoperative and postoperative Constant scores (33 ± 18 vs 20 ± 16 ; $P = .001$). There was no statistical difference between the type of lesion and number of tendons involved.

Duration of Symptoms Preoperatively

The median postoperative Constant score was 81.5 (IQR, 19.5) in patients with symptoms for ≥ 6 months at the

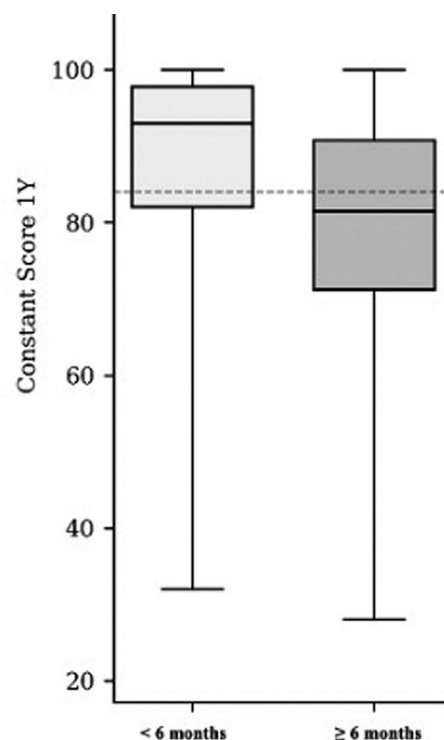


Figure 6. One-year follow-up median Constant score box plot by duration of symptoms.

time of surgery and 93.0 (IQR, 15.75) in patients with symptoms for < 6 months at the time of surgery ($P = .024$) (Figure 6, Table 3). Overall, 83% of patients with symptoms for < 6 months at the time of surgery had a traumatic tear. The preoperative Constant score was significantly lower in patients treated operatively < 6 months after the onset of symptoms (47 ± 18 vs 58 ± 12 points; $P = .032$). The increase from preoperative to postoperative Constant score was higher in patients with symptoms for < 6 months at the time of surgery than in patients with symptoms for ≥ 6 months at the time of surgery ($+39 \pm 28$ points vs $+21 \pm 12$ points; $P = .023$).

Complications

Nine (8.3%) patients had complications. Eight had postoperative stiffness that required a specific treatment (subacromial injections, rehabilitation for > 6 months). One patient needed a reoperation for symptomatic nonhealing. There was no case of infection in our series.

Multivariate Analysis

In multivariate analysis, shoulder trauma ($\beta = 8.03$; 95% CI, 2.6 to 13.46; $P = .0042$) was associated with higher postoperative Constant scores. Work-related tear ($\beta = -7.32$; 95% CI, -13.91 to -0.72 ; $P = .0301$) was associated with lower Constant scores at 1 year. Multiple-tendon tear

TABLE 4
Multivariate Analysis for Association With the Constant Score at the 1-Year Follow-up^a

	Coefficient (95% CI)	P Value
Traumatic tear	7.99 (2.69 to 13.29)	.00351
Level of work (reference: light manual worker)		
Heavy manual work	6.72 (−0.21 to 13.66)	.0572
Office work	5.07 (0.0137 to 10.12)	.0494
Work-related tear	−8.01 (−14.7 to −1.32)	.0194
Multiple-tendon tear	−3.03 (−8.12 to 2.07)	.241
Age at inclusion, y (risk for each 1-unit increase)	−0.247 (−0.594 to 0.101)	.163

^aBoldface type indicates statistical significance.

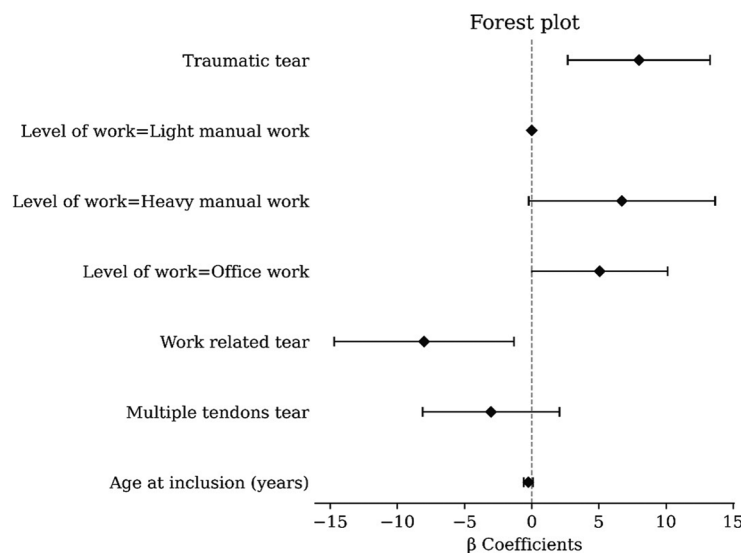


Figure 7. Forest plot of factors associated with Constant score at the 1-year follow-up.

($\beta = -3.25$; 95% CI, -8.38 to 1.88 ; $P = .2114$), age at inclusion (years) ($\beta = -0.33$; 95% CI, -0.69 to 0.04 ; $P = .0784$), and office work ($\beta = 2.56$; 95% CI, -2.33 to 7.46 ; $P = .3016$) were not associated with the postoperative Constant score at 1 year.

In multivariate analysis, patients with traumatic tears had, on average, an 8-point higher postoperative Constant score compared to those with degenerative tears ($P = .00351$), suggesting better recovery in this group. Conversely, patients with work-related tears experienced 8-point lower improvement ($P = .0194$), indicating a potential negative impact of occupational strain on functional recovery. Regarding occupational activity, patients in office-based jobs showed a 5-point greater improvement compared to light manual workers ($P = .0494$). In contrast, variables such as multiple-tendon tears (-3.03 points; $P = .241$) and age (-0.247 points per year; $P = .163$) did not show a statistically significant impact on postoperative outcomes (Table 4, Figure 7).

DISCUSSION

The main finding of this study was that patients aged ≤ 45 years with RCTs experienced good clinical outcomes, including improvement in pain, active forward elevation, active external rotation, strength, and Constant score. To our knowledge, this represents the largest series of RCR in patients ≤ 45 years of age. Our results are comparable to other studies published on this age group^{4,13,14,16,17,21}—even the study of Scanaliato et al²¹ involving active-duty military patients < 40 years of age. However, studies are difficult to compare precisely as either preoperative data are not specified or postoperative scores are different. Because of the design of our study, older patients were not included, so direct comparisons across age groups were not possible. Based on our hypothesis, younger patients may experience more symptoms preoperatively due to higher functional demands; indeed, in our series, patients engaged in heavy physical work

reported worse preoperative symptoms and showed less relative improvement postoperatively. In our series, most patients (82%) experienced a clinically statistically significant improvement.

In our series, traumatic tears had better improvement in Constant score than overuse tears in this age group. However, traumatic tears tend to be more symptomatic and be operated on earlier than overuse tears, which can have an influence on the results. Similar results have been published in the literature.¹⁶ On the contrary, Lin et al¹⁷ and Themessl et al,²⁴ in smaller cohorts, found no significant differences in outcomes between patients with traumatic tears and those with overuse tears. All studies in this age group emphasize that traumatic tears are more frequent than in the general population. However, the frequency varies. In the study of Lin et al,¹⁷ 60% of the patients in this age group had a traumatic event, compared to 41% in our series. However, the clinical outcome showed a similar mean Constant score to our series.

Type of occupation has an impact on the clinical outcome. Although all types of workers had a statistically significant difference in their postoperative Constant score at the final follow-up, office workers tended to have higher Constant scores and higher improvement compared to their preoperative state. Lower Constant scores in light manual workers can be at least partially explained by female sex being highly represented in this group, with lower strength in general compared to a male cohort. In our series, work compensation status had a negative influence on postoperative outcome after RCR, as has been previously shown in the literature.^{5,7,12,18,19,24,26}

Duration of symptoms also has an impact on postoperative results. In our study, higher postoperative Constant scores were found in patients operated on within 6 months after onset of symptoms compared to patients waiting for longer. Traumatic tears tend to be operated on sooner than degenerative tears. However, more symptomatic patients tend to come for a clinical evaluation earlier and tend to be operated on earlier as well.³ Hantes et al¹² showed that traumatic tears experienced better clinical improvement if treated earlier. Similar results in a general population were shown by Petersen and Murphy.²⁰ Whereas meta-analysis did not show duration of symptoms as a prognostic factor,^{8,26} Godenèche et al¹⁰ demonstrated that even stage 1 fatty infiltration has a negative influence on the postoperative outcome at 10 years. Indeed, in their study, the risk of retear of the supraspinatus varied depending on the degree of fatty infiltration: 10% for stage 0, 22% for stage 1, and 31% for stage 2. This emphasizes the importance of treating this age group of patients early.

We were surprised to observe such a long delay between onset of symptoms and surgical treatment. This is likely explained by the predominance of nontraumatic tears in our series, where surgery was generally indicated only after the failure of extended nonoperative treatment. In most studies, operative treatment was considered only after 3 to 6 months of nonoperative treatment even in this age group. Moreover, 43% of patients in our series had a cortisone injection before surgery, despite the young

age of these patients. This calls into question the treatment recommendations of RCT in this age group. Indeed, recommendations of RCT should probably be adapted to age group, whereas public health recommendations in France (Haute Autorité de Santé), for example, do not make any distinctions. Further studies are needed to evaluate the best timing for surgery in this group of patients.

The main strength of this study is the number of patients included, which is the largest series so far. Compared to other studies published, we describe preoperative and postoperative outcomes with complete clinical parameters, including strength.

Our limitations include a relatively short minimum follow-up for some cases. In our series, 71% of patients had a follow-up of ≥ 13 months. While a longer follow-up was intended, retention was challenging given the young and active nature of our population, as many patients with good outcomes at 1 year declined further evaluation despite multiple invitations. Nevertheless, the mean follow-up exceeds 2 years. In this study, we focused on a relatively underrepresented population—patients ≤ 45 years of age undergoing RCR. While the primary aim was descriptive, we performed adjusted analyses that accounted for key tear characteristics, including traumatic versus nontraumatic origin. Although we were not able to adjust for symptom duration due to inconsistencies in the available retrospective data, our findings still provide valuable insights into outcomes in this younger demographic. Another important limitation is the absence of systematic postoperative imaging to assess tendon integrity and healing rates. While clinical outcomes provide meaningful information, structural healing may also influence long-term results and should be addressed in future studies.

The inclusion of both traumatic and atraumatic tears and the size of our cohort enhance the generalizability of the results. We recognize that prognostic factors such as time to surgery are likely important in influencing outcomes and merit further prospective investigation. Future studies with standardized data collection may be able to better address the impact of such variables in this population. Multiple centers were involved, but all surgeries were performed by senior shoulder surgeons. For this reason, different surgical techniques were used, as described above. Different types of surgery and immobilization were accepted as there is no proof in the literature that one technique outperforms another.²² We consider the impact of this limitation to be minimal. Moreover, as different scores and different age cutoffs are used in the literature, it is difficult to compare studies involving younger patients.

CONCLUSION

Patients ≤ 45 years of age with RCT generally have a good clinical outcome with improvement of pain and Constant score after arthroscopic repair. Degenerative tears and manual work are associated with poorer but acceptable outcomes.

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